

CoinHarbour

Protocol Whitepaper

V1

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1. Introduction

a. Abstract

CoinHarbour is a permissionless on-chain protocol for creating, issuing, redeeming, and rebalancing a fungible, collateralized set of tokens (funds). Users looking for passive exposure to diversified assets can easily subscribe to these funds, which are governed and managed by their deployers.

b. Core Philosophy

Complex systems across various domains rely on abstraction to improve accessibility and efficiency. In software engineering, abstraction hides unnecessary details, allowing users to interact with higher-level functionalities without managing low-level complexities.

TradFi follows a similar approach—mutual funds pool diverse assets into a single investment vehicle, while ETFs enable exposure to entire market sectors through a single trade. These mechanisms streamline interactions, lower cognitive load, and enhance usability.

c. Problem Statement

As Web3 adoption grows, tokenized assets are becoming the foundation of digital economies, spanning DeFi, gaming, governance, and RWAs. However, for users, managing these assets remains complex and fragmented. Each token exists in isolation, requiring separate transactions for transfers, swaps, and portfolio adjustments. This leads to high gas fees, manual effort, and an overwhelming UX.

Without a native way to structure and manage diversified investments, users must constantly monitor market movements, rebalance holdings, and execute trades manually, making passive exposure nearly impossible. Wallets and interfaces are not designed for passive multi-asset management, forcing users to juggle multiple platforms and track performance manually. A scalable solution is needed—one that abstracts away complexity, enables effortless portfolio management, and allows users to gain diversified exposure without the friction of managing each token separately.

2. CoinHarbour Protocol

a. Overview

Our protocol is a fully on-chain, permissionless infrastructure for deploying, issuing, redeeming, and rebalancing fungible, collateralized tokenized assets known as funds. Since it is smart contract based, it eliminates the need for intermediaries, ensuring trustless execution, verifiable transparency, and immutable settlement.

At its foundation, the protocol utilizes composable smart contract primitives to manage tokenized asset pools, enabling deterministic issuance and redemption mechanisms. These pools are governed by self-executing logic that enforces asset composition, collateralization rules, and rebalancing conditions, all while maintaining cryptographic auditability.

By abstracting away the complexities of multi-asset management, the protocol enhances liquidity efficiency, modular interoperability, and permissionless innovation. Participants can engage with tokenized compositions in a trust-minimized manner, leveraging the protocol as a foundational layer for decentralized asset coordination.

b. Key Differentiators

i. Trustless & on-chain – Smart contracts autonomously manage fund creation, rebalancing, and redemptions, ensuring verifiable execution without intermediaries.
ii. Permissionless & Composable – Any user can deploy and interact with tokenized asset pools, seamlessly integrating with DeFi ecosystems.
iii. Interoperable Fund Tokens – Issued fund tokens follow widely adopted token standards, enabling seamless compatibility across decentralized exchanges, lending protocols, and other DeFi applications.
iv. Immutable & Transparent – All fund operations are executed and recorded on-chain, ensuring full audibility and eliminating reliance on opaque asset management.
v. Automated & Capital Efficient – Algorithmic rebalancing and optimized execution enhance capital efficiency and reduce operational overhead.
vi. Gas-Optimized – Designed for minimal transaction costs, ensuring efficient and scalable fund management on-chain.

c. CoinHarbour Fund

i. Overview

CoinHarbour Funds are on-chain, permissionless vault tokens that algorithmically represent a composable basket of underlying assets. These tokens are fully collateralized and can be trustlessly minted or burnt via smart contracts, ensuring immutable transparency and verifiability. Each fund token is a liquid, transferable asset, inheriting interoperability across DeFi primitives such as AMMs, lending markets, and yield protocols.

The valuation of a fund token is directly backed by its underlying collateral, with its price dynamically reflecting the aggregated value of all assets it holds. As market conditions shift, the token's value fluctuates accordingly, maintaining a transparent, on-chain representation of its real-time worth.

ii. Incentivisation

CoinHarbour Funds align incentives between fund creators and users through a trustless fee mechanism. Fund creators can set a flat withdrawal fee as a percentage of realized profits, ensuring they earn only when users do, reinforcing the protocol's commitment to equitable value distribution. This performance-based model discourages unnecessary fees and ensures that fund managers are incentivized to deploy optimal strategies while users retain confidence in the integrity of their investments.

It can be mathematically expressed as:

$$\text{Fee} = \begin{cases} \text{profit} \times \text{feeRate}, & \text{if profit} > 0 \\ 0, & \text{if profit} \leq 0 \end{cases}$$

Where

$$\text{profit} = \text{withdrawalAmount} - \text{initialDeposit}$$

$$\text{feeRate} = \text{CreatorDefined percentage fee on profit}$$

iii. Use Cases

- **ETFs & Index Funds**
  Tokenized index funds and ETFs enable passive exposure to a diversified basket of assets, allowing users to invest in broad market sentiments.

- **Mutual Funds & Theme-Based Baskets**
  Funds can focus on specific investment themes like DeFi, gaming, or ecosystem-based strategies, offering targeted exposure.

- **Liquidity Vaults**
  Users can deposit assets into actively managed vaults that generate yield through lending, staking, or arbitrage strategies while maintaining liquidity via tokenized fund shares.

- **Yield Aggregators**
  Funds autonomously allocate capital across multiple DeFi protocols to optimize returns, auto-compound earnings, and minimize manual intervention.

- **Social Trading Funds**
  Experienced traders can create fund tokens representing their strategies, allowing others to invest in their portfolios with performance-based incentivisation.

- **AI Agents & Trading Bots**
  Automated funds powered by AI models or algorithmic trading bots execute predefined strategies, optimizing buy/sell decisions and market timing with no human intervention.

- **Treasury Management**
  DAOs and protocols can diversify their treasuries by holding fund tokens, enabling strategic allocation while maintaining liquidity and reducing exposure risk.

- **Synthetic Asset Bundles**
  Funds can bundle tokenized real-world assets (RWAs), synthetic stocks, commodities, or yield-bearing instruments, allowing users to gain exposure to off-chain markets in a decentralized manner.

iv. Fund Mechanics

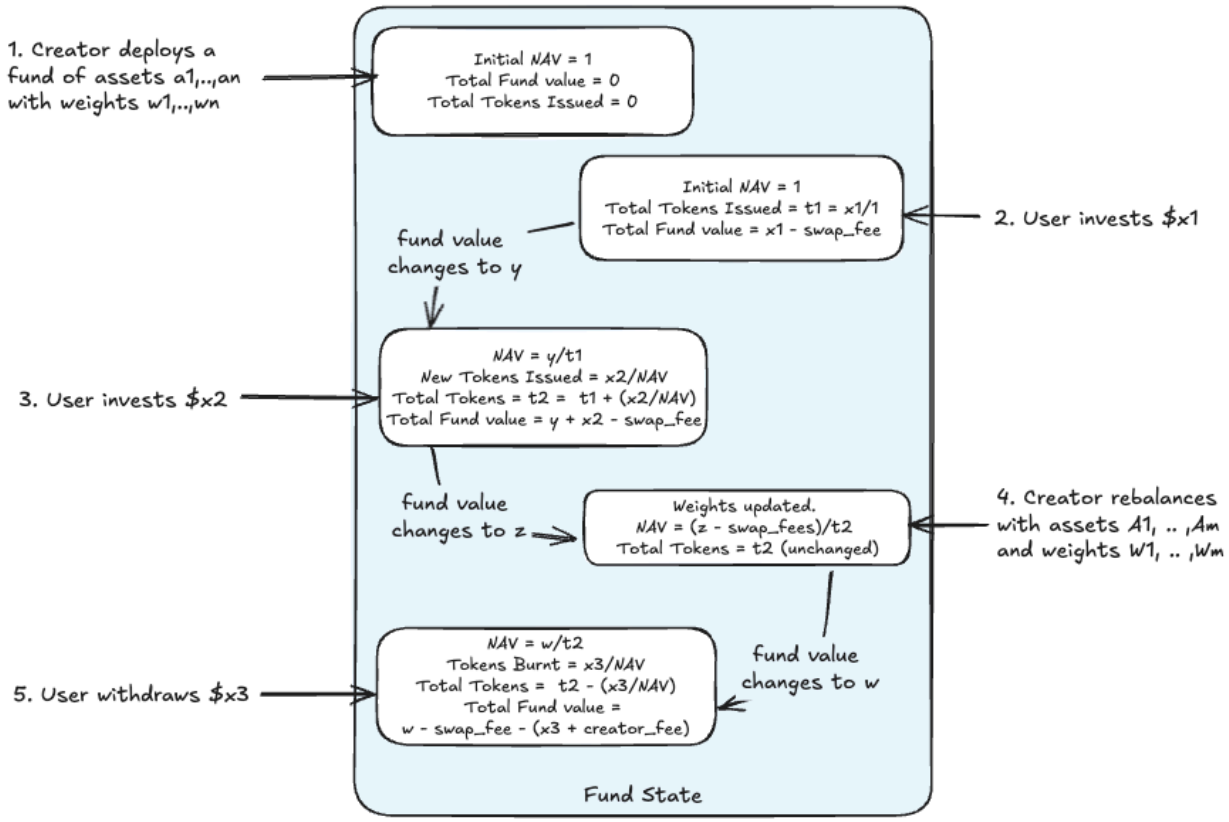
Fund tokens dynamically derive their value from the underlying assets they represent. These assets are collateralized, and their price fluctuations directly impact the fund's valuation. Three key metrics govern fund mechanics:

- **Net Asset Value (NAV):** The price per fund token, calculated as

$$\text{NAV} = \frac{\text{Total Fund Value}}{\text{Total Fund Supply}}$$

- **Total Fund Value:** The aggregate market value of all assets held by the fund.
- **Fund Token Supply:** The total number of issued fund tokens, increasing with deposits and decreasing with withdrawals.

Note: Each fund token maintains a proportional claim over the vault's assets, ensuring price discovery and transparency through smart contracts.



Note:- every state will have a pre-transaction NAV, on which the #tokens to mint/burn will be decided and a post-transaction NAV, accounting the swap/gas fees. At all stages:- Post-Transaction NAV = Pre-Transaction NAV - fees/total_tokens_issued

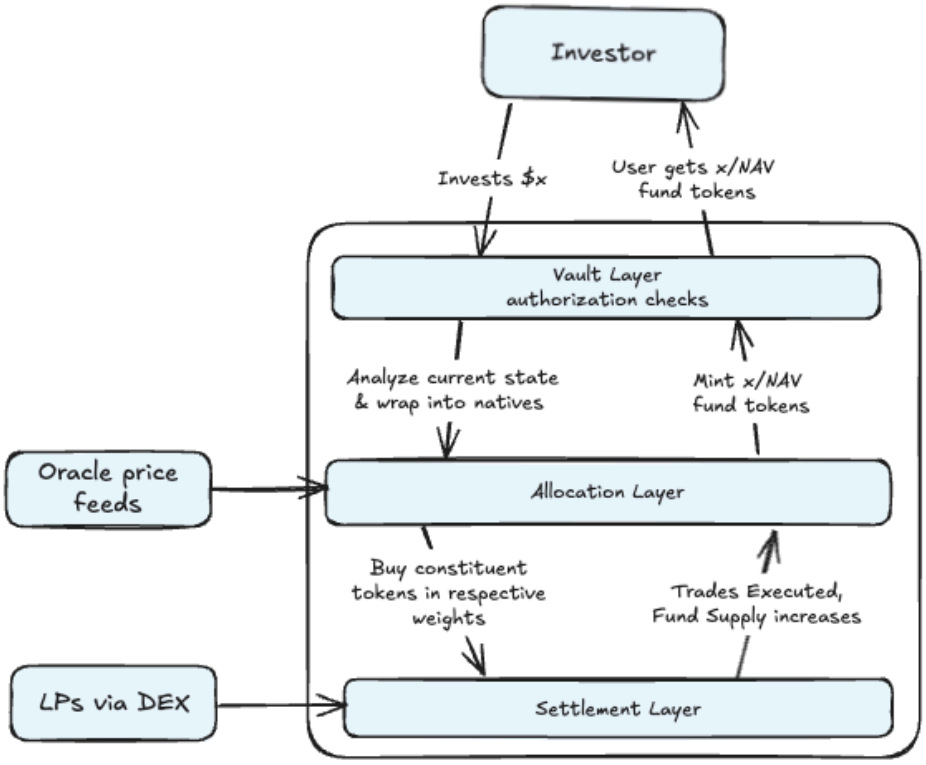
1. Invest

When a user deposits capital, their funds are programmatically allocated to the fund's asset composition. This happens in the following steps:

a. **Minting Fund Tokens**: Based on the latest fund state, the user is minted a corresponding amount of fund tokens at the current NAV.
b. **Swap & Allocation**: The deposit amount is swapped proportionally into the fund's collateralized assets, maintaining the pre-set weights.
c. **Updated Fund State**: The fund's total value increases, and its token supply grows proportionally.

Example

1. User deposits $10,000 into the vault, with current NAV = 1.
2. New 10,000 (10000 ÷ 1) fund tokens minted and issued to the user.
3. Vault executes swaps to collateral assets in the current ratio.
4. Assuming $2 swap fee (gas) is burnt during swaps.
5. Output assets worth $9,998 get added to the vault, leaving NAV = (9998 ÷ 10000) = 0.998



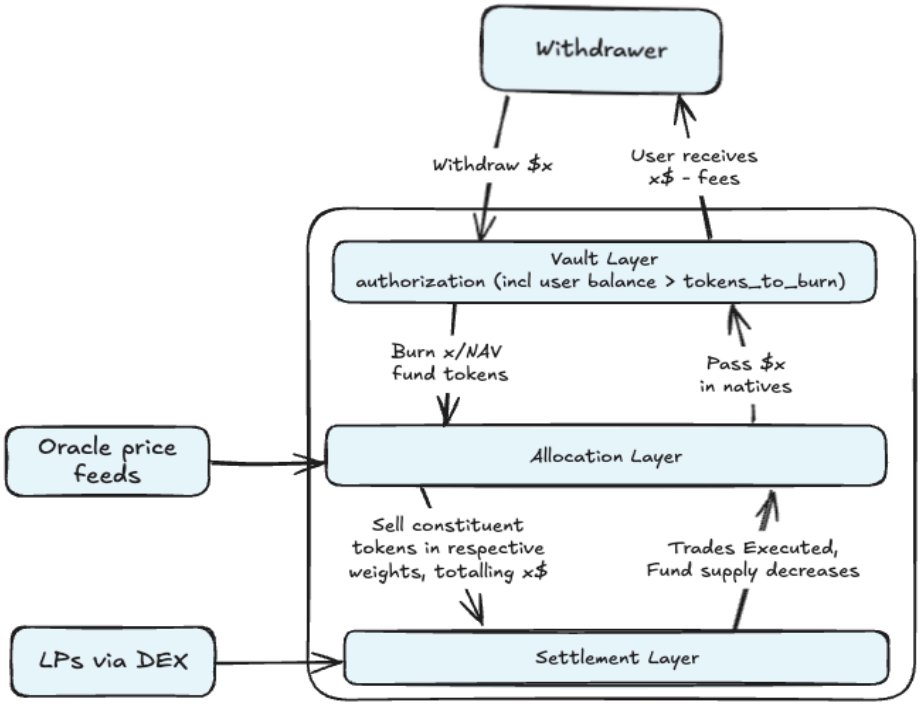
2. Withdraw

Withdrawals follow an inverse process, ensuring proportional ownership is preserved:

a. **Burning Fund Tokens**: The user returns their fund tokens, which are permanently removed from circulation.
b. **Asset Swap & Distribution**: The equivalent value of collateralized assets is sold for stablecoins (or other withdrawal currencies).
c. **Final Payout**: The user receives their withdrawal amount, post the creator fee deduction, and the fund supply shrinks accordingly.

Example

1. Assume that the value of the collateral assets has increased, and the initial $9,998 worth of assets is now worth $15,000.
2. Total fund value is now $15,000, with new NAV = 1.5 (15000 ÷ 10000).
3. User requests $8,000 withdrawal, burning 5,333.33 fund tokens (8,000 ÷ 1.5). User must hold > 5333.33 tokens.
4. Assuming a $3 swap fee while selling, thereby receiving stables worth $7,997.
5. User receives $7,997, subject to creator fee.



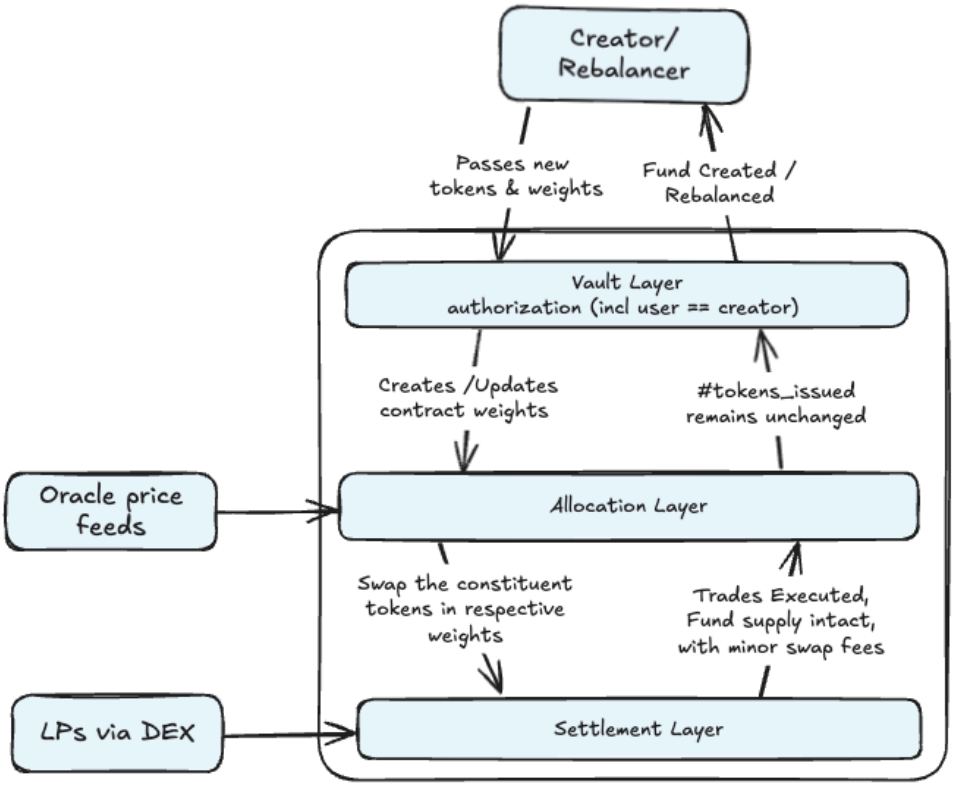
3. Rebalance

Rebalancing ensures the fund maintains its intended asset composition. It can be automated via smart contracts or executed manually by fund managers. The process involves:

a. **Asset Swaps**: Some assets are sold while others are acquired to adjust allocation percentages.
b. **Maintaining NAV**: The fund supply remains constant, but due to swap fees, there is a slight decrease in total fund value.
c. **Updated Portfolio**: The new composition takes effect, aligning with market trends or strategy updates.

Example

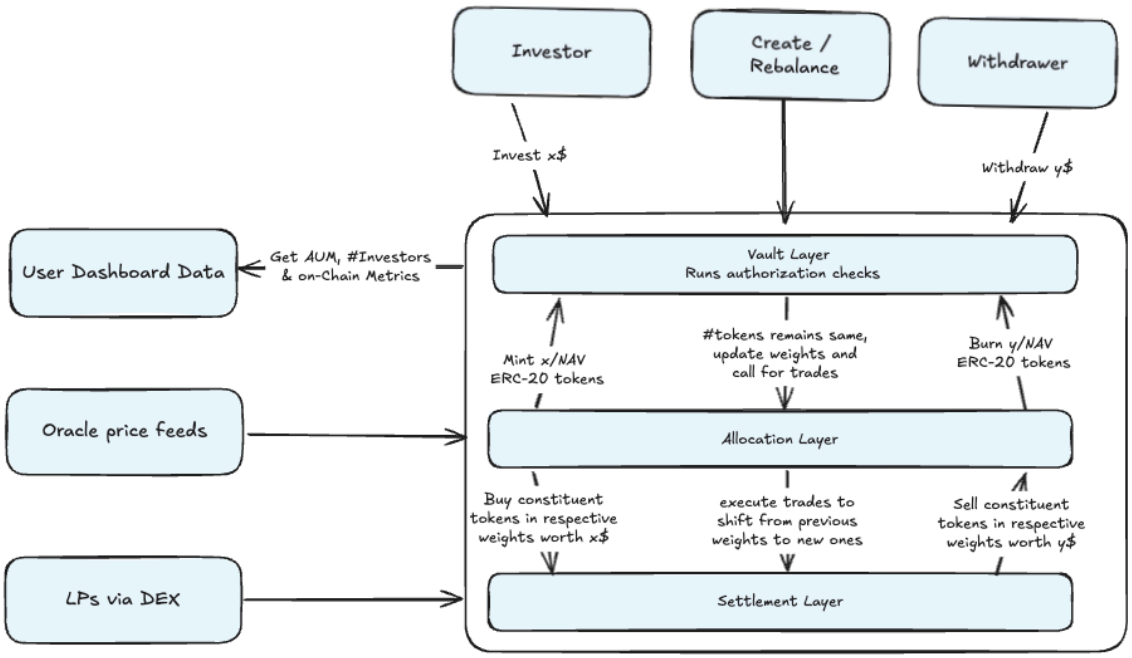
1. Initial Allocation: 40% BTC, 30% ETH, 30% SOL.
2. Market Shift: BTC rises, shifting weights to 45% BTC, 28% ETH, 27% SOL.
3. Adjustment: Rebalance call for 50% BTC, 25% ETH, 25% SOL.
4. Swaps: ETH & SOL swapped for BTC, incurring $5 fees.
5. NAV Impact: Drops slightly from $1.50 → $1.498.
6. Final State: Portfolio realigned, strategy maintained.



3. Technical Implementation

a. Overview

The fund operates entirely on-chain, leveraging smart contracts to handle deposits, withdrawals, rebalancing, and asset management. These contracts interact with decentralized exchanges (DEXs) and price oracles to maintain the fund's asset composition while ensuring transparency and security. The fund also exposes various pure view methods for reading on-chain metadata.



b. Components

i. Vault Layer

1. Handles deposits and withdrawals of assets into the fund.
2. Minting and burning of tokens to reflect fund share ownership.
3. Runs authentication checks including user balance, creator rights.

ii. Allocation Layer

1. Responsible for maintaining the fund's target asset allocation.
2. Executes rebalancing by swapping assets in predefined proportions.
3. Fetches real-time price data from oracles to assure optimal orders.

iii. Settlement Layer

1. Interfaces with on-chain liquidity sources (DEXs) for executing trades.
2. Ensures efficient execution and minimizes slippage.
3. Listens for periodic or event-driven rebalancing.

4. Protocol Sustainability

The protocol ensures long-term sustainability through economic incentives and efficient on-chain operations. These mechanisms align the interests of investors and developers while maintaining decentralization and security.

a. **Revenue Model**:
The protocol generates revenue exclusively through withdrawal fees. A minimal fee is charged when withdrawals are made. This ensures protocol sustainability while keeping costs low for users.

b. **Governance & Decentralization**:

1. Progressive Decentralization – Governance will transition from core team control to a DAO-driven model where token holders participate in protocol decisions.
2. DAO Treasury Management – Fees collected from transactions are allocated to development, security audits, and ecosystem expansion to maintain continuous innovation.

c. Security & Risk Management:

1. Automated Risk Controls – Integrated safeguards prevent excessive leverage, liquidation cascades, and unintended fund exposure.
2. Smart Contract Audits – Regular audits and bug bounty programs enhance security and reliability.
3. Failsafe Mechanisms – Circuit breakers and time-locked governance proposals protect against unexpected failures and malicious actions.

By focusing on efficient fee-based revenue, decentralized governance, and strong security, the protocol remains self-sustaining and resilient over the long term.

5. Launch Economics

a. **Overview**

coinHarbour.xyz is designed to be fully on-chain, with strategy execution, fund management, and governance occurring entirely within smart contracts. Only simulation-based components such as backtesting, strategy evaluation, and user data storage are processed off-chain to ensure efficiency and scalability.

b. **Verticals**

i. **Fund baskets**

These are manually rebalanced tokenized asset pools built on the protocol's funds. They enable users to deploy structured on-chain portfolios with predefined asset compositions, enforced by smart contracts. They rely on governance or fund managers to trigger rebalancing events, adjusting asset compositions through the trade execution layer interactions with DEXs all abstracted away. This module is ideal for structured, thematic, or discretionary portfolio strategies that require flexible manual oversight.

ii. **AI & Algorithmic Trading Agents**

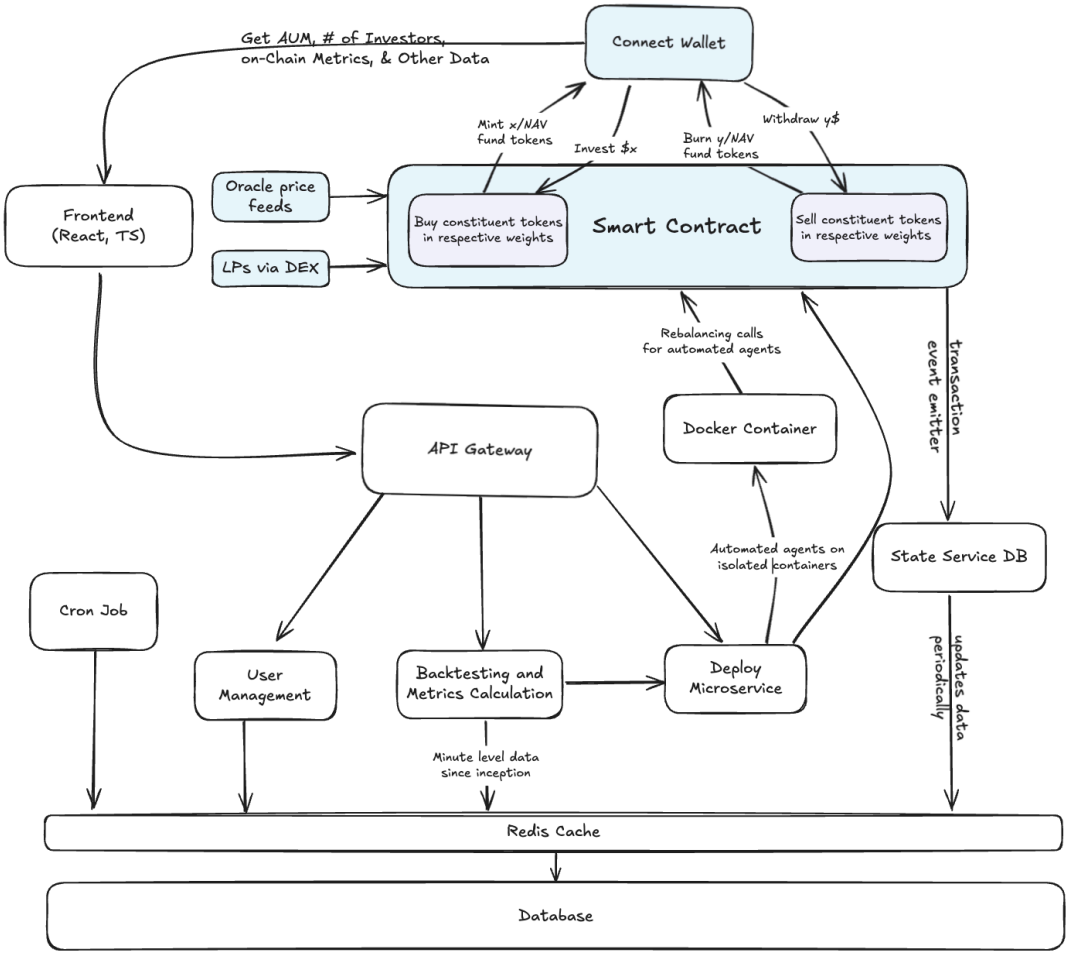
These are algorithmically or AI rebalanced tokenized asset pools, also leveraging the protocol's fund for automated strategy execution. They continuously adjust asset allocations based on predefined logic, interacting with price oracles to fetch real-time market data and executing trades. By automating trades, they ensure capital efficiency and seamless portfolio adjustments without manual intervention. This module is designed for fully decentralized, self-sustaining asset management, enabling passive exposure to dynamically managed on-chain portfolios.

c. **Features**

i. Truly On-Chain → Decentralized execution with cross-chain interoperability.
ii. Advanced Backtesting → Minute-level simulations with 7+ years of historical data.
iii. Reliable Market Data → Aggregated from decentralized oracle networks.
iv. Flexible & Accessible → No lock-in, no minimum investment, full liquidity control.

d. Technical Architecture

While the core trading execution and fund management remain on-chain, the off-chain infrastructure enhances performance and user experience.



1. **Frontend & Wallets**: React-based Web UI with Wallet integrations.
2. **Backend**: Go microservices power user management, simulation & backtesting, and off-chain data storage.
3. **AI, Algo Trading & Analytics**: Python microservice handles bots, fund rebalancing, AI agents, and portfolio metrics..
4. **Live & Historical Data**: Real-time OHLCV data triggers on-chain rebalancing, while Parquet storage enables efficient backtesting.
5. **Data Storage**: PostgreSQL stores fund configurations, metadata, backtesting reports, and rebalance histories, with Redis cache.
6. **Deployment**: Isolated Docker containers execute trading strategies, Traefik manages routing, and a distributed architecture prevents interference between users.

6. Conclusion

a. Summary

CoinHarbour is a fully on-chain, permissionless protocol for deploying, issuing, redeeming, and rebalancing fungible, collateralized funds. Built entirely on smart contracts, it eliminates intermediaries, ensuring trustless execution, transparent settlement, and immutable asset management.

At its core, the protocol utilizes composable smart contract primitives to govern tokenized asset pools, enforcing deterministic issuance, redemption, and rebalancing while maintaining cryptographic auditability. By abstracting multi-asset management complexities, it enhances liquidity efficiency and interoperability within DeFi.

CoinHarbour Fund tokens algorithmically represent collateralized baskets of assets, maintaining transparent and verifiable backing. These liquid, transferrable tokens integrate seamlessly across DeFi protocols, inheriting interoperability with AMMs, lending markets, and yield platforms, ensuring efficient and decentralized asset coordination.

b. Participate

Look, we get it. Crypto investing shouldn't be complicated. That's exactly why we built CoinHarbour differently. No BS, no fancy buzzwords, just a platform that actually lets regular people invest in crypto without needing a PhD in degen blockchain tech. If you're tired of watching from the sidelines or getting rekt by trying to do everything yourself, we've got you covered.

Ready to see what we've got? Hit us up at *coinharbour.xyz*.

Disclaimer: This whitepaper is for informational purposes only and does not constitute financial advice. Cryptocurrency investments carry inherent risks. Please conduct your own research before making any investment decisions.